

Damien Robert

Deep Learning for Remote Sensing & Environment



✉ damien.robert@uzh.ch · 🌐 drprojects.github.io · 📄 drprojects · 🆔 0000-0003-0983-7053 · 🎓 Scholar · 🌐 damienrobert91

I am a postdoctoral researcher in the [EcoVision](#) lab at the [University of Zurich](#), where I collaborate with [Jan Dirk Wegner](#) to design deep learning methods for remote sensing and environmental applications. Before joining UZH, I completed my PhD on “*Efficient Learning on Large-Scale 3D Point Clouds*” at [IGN](#) and [ENGIE](#), under the supervision of [Loic Landrieu](#) and [Bruno Vallet](#).

3D point clouds · vegetation mapping · species distribution · remote sensing · efficient ML

■ POSITIONS

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|-----------|--|
| 2024–now | Postdoctoral Researcher
EcoVision lab, University of Zurich · Zurich, Switzerland
Deep learning for remote sensing and environment
PI: Jan Dirk Wegner |
| 2020–2024 | PhD Student
LASTIG, IGN / ENGIE Lab CRIGEN · Paris, France
Efficient learning on large-scale 3D point clouds
Advisors: Loic Landrieu, Bruno Vallet |
| 2017–2020 | R&D Engineer
SIRADEL, ENGIE / ENGIE · Rennes, France
Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data |
| 2017 | Co-Founder
Inspirama · Rennes, France
Website gathering book recommendations from inspiring people |
| 2015 | R&D Intern
Dassault Systèmes · Providence, RI, USA
Dimensionality reduction and dynamic system modeling |
| 2014 | R&D Intern
Dassault Systèmes · Providence, RI, USA
UX design |
| 2014 | R&D Intern
Dassault Systèmes · Providence, RI, USA
UX design |

■ EDUCATION

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| 2020–2024 | PhD, Computer Science
Université Gustave Eiffel · Paris, France
Efficient Learning on Large-Scale 3D Point Clouds |
| 2022 | Summer School
ICVSS · Sicily, Italy
Computer vision courses by world-renowned experts in academia and industry |
| 2017 | AI Fall School
CNRS · Lyon, France
Multi-disciplinary course for AI students and researchers |
| 2011–2015 | MSc, Engineering
École Centrale de Lyon · Lyon, France
Mathematics, Computer Science, Mechanics, Signal Processing, Automation |

■ AWARDS

2026	Outstanding Reviewer · CVPR
2025	PhD Award accessit · AFRIF French Association for Pattern Recognition and Interpretation
2024	Outstanding Reviewer · ECCV
2022	Best Paper Finalist · CVPR Top 0.4% of submissions

■ PUBLICATIONS

12 papers · 487 citations · h-index 7 · Google Scholar, Sep 2026

2026	Toward a Foundation Model for Forest Point Clouds Y. Yue, S. Puliti, D. Robert , A. Topaloglu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler <i>Under review</i> From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps G. Sialelli, R. Young, Y. Jiang, ..., D. Robert , et al. (13 authors) <i>Under review</i> Bias Aware Benchmark for Species Distribution Modeling E. Arens, N. van Tiel, R. Zbinden, C. Vanalli, D. Robert , L. Drees, B. Kellenberger, L. Pellissier, J. D. Wegner, D. Tuia World Biodiversity Forum LitePT: Lighter Yet Stronger Point Transformer Y. Yue, D. Robert , J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler CVPR Highlight EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation L. Geist, L. Landrieu, D. Robert ICRA FORMSpot: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series M. Schwartz, F. Fogel, N. Besic, D. Robert , L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciais Remote Sensing of Environment
2025	MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation K. Karaman, V. S. F. Garnot, D. Robert , M. J. Santos, J. D. Wegner PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology E. Plekhanova, D. Robert , J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann CVPR workshop EarthVision GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance K. Karaman, Y. Jiang, D. Robert , V. S. F. Garnot, M. J. Santos, J. D. Wegner ISPRS Geospatial Week Oral Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks J. Dollinger, D. Robert , E. Plekhanova, L. Drees, J. D. Wegner ICLR workshop Tackling Climate Change with Machine Learning Lossy Neural Compression for Geospatial Analytics: A Review C. Gomes, I. Wittmann, D. Robert , et al. (27 authors) IEEE Geoscience and Remote Sensing Magazine
2024	Efficient Learning on Large-Scale 3D Point Clouds D. Robert PhD Thesis, Université Gustave Eiffel AFRIF PhD award accessit Jury: Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loic Landrieu, Bruno Vallet Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering D. Robert , H. Raguet, L. Landrieu 3DV Oral · top 5.3% of submissions
2023	Efficient 3D Semantic Segmentation with Superpoint Transformer D. Robert , H. Raguet, L. Landrieu

ICCV · 26.8% acceptance rate

2022

Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation

D. Robert, B. Vallet, L. Landrieu

CVPR **Best paper finalist** **Oral** · top 0.4% of submissions

■ COLLABORATION & SUPERVISION

Collaborators	Konrad Schindler ^{ETH} · Christian Rupprecht ^{Oxford} · Jan Dirk Wegner ^{UZH} · Loic Landrieu ^{ENPC} · Vivien Sainte Fare Garnot ^{UZH} · Lukas Drees ^{UZH} · Martin Schwartz ^{LSCE} · Elena Plekhanova ^{WSL}
PhD Students	Emilia Arens ^{UZH} with Jan Dirk Wegner · Johannes Dollinger ^{UZH} with Jan Dirk Wegner · Kaan Karaman ^{UZH} with Jan Dirk Wegner · Louis Geist ^{ENPC} with Loic Landrieu · Yuanwen Yue ^{ETH} with Konrad Schindler and Christian Rupprecht
Master Students	Valerio Schelbert ^{WSL} · Jackson Sunny ^{École Polytechnique} with SAMP
Advising	UpCircle AI ^{Switzerland} · École des Sciences Criminelles de Lausanne ^{Switzerland} · SAMP AI ^{France} · GEODIGITAL ^{US}

■ REVIEWING, CHAIRING & ORGANISING

 Reviewer  Area chair  Organising committee

2027	 CVPR workshop EarthVision
2026	 ECCV ·  CVPR
2025	 CVPR ·  CVPR workshop EarthVision
2024	 ECCV ·  ISPRS Journal of Photogrammetry and Remote Sensing ·  ICLR workshop ML4RS ·  CVPR workshop EarthVision
2023	 CVPR ·  CVPR workshop EarthVision
2022	 ISPRS Journal of Photogrammetry and Remote Sensing ·  CVPR workshop EarthVision

■ TEACHING

2026	Transformer · ^{UZH} · Lecture, Master 2
2025	AI4Good group projects · ^{UZH} · Labs, Master 2
2025	Transformer · ^{UZH} · Lecture, Master 2
2024	Transformer, NeRFs, and Diffusion · ^{UZH} · Lecture and labs, Master 2
2023	Deep Learning for Remote Sensing · ^{ENSG-IGN} · Lecture, Master 2
2022	3D Deep Learning for Remote Sensing · ^{ISPRS'22} · Tutorial, Researchers
2022	3D Deep Learning, Torch-Points3D & DeepViewAgg · ^{ENGIE CRIGEN lab} · Tutorial, Researchers
2022	Deep Learning for Remote Sensing · ^{ENSG-IGN} · Lecture, Master 2
2020	Deep Learning for Computer Vision · ^{École Polytechnique} · Lecture, Master 1

■ OPEN-SOURCE SOFTWARE

★ 1055	 136	Superpoint Transformer ·  drprojects/superpoint_transformer
★ 406	 45	LitePT ·  prs-eth/LitePT
★ 237	 24	DeepViewAgg ·  drprojects/DeepViewAgg
★ 74	 8	Point Geometric Features ·  drprojects/point_geometric_features
★ 53	 3	NoRA ·  drprojects/nora
★ 15	 0	torch-graph-components ·  drprojects/torch-graph-components

■ SELECTED TALKS

 Invited  Conference  Poster  Tutorial  Interview

2026	 ESA-NASA workshop on AI Foundation Model for EO · Building Earth Observation Embedding Workflows with TerraTorch · Virtual
	 ESA Φ -lab · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy

2025	🗨️ MPI of Animal Behavior · Research at EcoVision lab · Zurich, Switzerland 🗨️ Global Forest Watch · Geospatial foundation models for forest disturbance monitoring · Virtual 🗨️ NatureFinance · Research at EcoVision lab · Virtual 🗨️ NVIDIA Strategic Research Engagement · Superpoint Transformer and EcoVision research · Zurich, Switzerland 🗨️ CNES COMET TSI · Remote Sensing for Species Distribution Modeling · Virtual
2024	🗣️ 3D Data Academy · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube 🗨️ Google DeepMind · Research at EcoVision lab · Zurich, Switzerland 🎤 3DV'24 · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland
2023	🗨️ École des Ponts, IMAGINE lab · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France 🗨️ National Land Survey of Finland (NLS) · IGN's research on Large-Scale 2D and 3D Learning · Paris, France 🗨️ ETH Zürich, Computer Vision and Geometry lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland 🗨️ ETH Zürich, Photogrammetry and Remote Sensing lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland 🗨️ Valeo.ai · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France
2022	🗣️ CV News · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US 🎤 CVPR'22 · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US
45 talks given since 2021 — full list at drprojects.github.io/talks .	

■ SKILLS & LANGUAGES

Research	3D & geometry Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning Efficient & scalable learning Efficient deep learning · Scalable model design · EO data representation and compression Learning paradigms Multimodal learning · Self-supervised learning · Representation learning Environment & Earth observation Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology			
Tools	 Python	 PyTorch	 PyTorch Lightning	 PyTorch Geometric
	 C++	 Hydra	 Weights & Biases	 Git
	 Linux	SLURM	 scikit-learn	 Plotly
	 Blender	LaTeX		
Languages	French Native	English Fluent	Spanish Intermediate	German Beginner
Abroad	Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015			
Interests	Nature & forests · Boxing · Mountaineering · Skateboarding · Guitar · Painting · Reading · Board games			